



To the Editor of the
Environmental Modelling & Software

Dear Editor,

We are glad to resubmit the deeply revised version of our manuscript, now entitled “**Shoreline compactness predicts Sentinel-1 water-area monitoring reliability across reservoirs worldwide**”, by Cicero Martins Jr., Fulvio Capodici, Mauro De Marchis, and Giuseppe Ciraolo, submitted for consideration as a new submission to *Environmental Modelling & Software*.

The corresponding author is MSc. Cicero MARTINS Jr.

The corresponding author affirms that:

- All authors of this research paper have directly participated in the planning, execution, or analysis of this study;
- All authors of this paper have read and approved the version submitted;
- The contents of this manuscript have not been copyrighted or published previously;
- The contents of this manuscript are not now under consideration for publication elsewhere.

As instructed in your decision letter of 8 June 2026, we reference here the prior manuscript number, **ENVSOFT-D-26-00848** (“Monitoring Reservoir Surface and Storage Dynamics Using Sentinel-1 SAR and Machine Learning in Google Earth Engine”).

Below, we detail point by point how the manuscript has changed in response to your comments and those of both reviewers. The paper has been substantially reframed rather than incrementally patched, so we first summarize the reframing before addressing each comment individually.

Summary of the reframing

Both reviewers and the handling editor converged on three concerns: insufficient novelty relative to prior SAR-plus-machine-learning studies, an unsubstantiated scalability claim, and an experimental setup validated on too few reservoirs to support its conclusions. Rather than defend the original framing, we changed what the paper is about.

- **Novelty.** The manuscript’s central claim is no longer that a GEE workflow using VV/VH Sentinel-1 and an SVM can monitor Sicilian reservoirs. It is that a reservoir’s shoreline **area-to-perimeter ratio (A/P)** — a length-scale computable from global databases before any image is processed — predicts, a priori, the geometric ceiling on SAR classification skill (Introduction, Section 1; Section 3.6; Results, Section 4.1; Discussion, Section 5.2). This claim was already at the previous paper, and it is the direction Reviewer 1 identified as the paper’s most interesting, under-exploited thread (their General Comment 3).
- **Scalability.** All manual training-sample delineation has been removed. Both classifiers are now trained automatically from global JRC Global Surface Water and ESA WorldCover masks (Section 3.3), and this fully automated pipeline is tested on **62 reference-quality-screened reservoirs across 33 countries and five continents** (Section 2.1), not on a single region.



- **Experimental setup and scope.** The manuscript no longer attempts to convert surface area to storage volume. That conversion depended on area–elevation–volume (AEV) curves built from historical design surveys, and, as Reviewer 1 rightly noted (specific comment 12), our attempt to justify replacing them while still relying on them was internally inconsistent. We now scope the paper to what it can support rigorously, the SAR-derived surface area itself, and leave the AEV/volume problem to future work building on this result (Discussion, Section 5.4). This change resolves the majority of the Results and Discussion issues raised by both reviewers (Reviewer 1 comments 12, 17–20; Reviewer 2 items E.4, E.5, F), since the disputed material no longer appears.
- **Validation.** The four Sicilian reservoirs are retained, but now as a high-resolution (3 m PlanetScope) near-truth check layered on top of the 62-reservoir global comparison against the JRC Global Surface Water record (Sections 2, 3.7), rather than as the study’s only validation.
- **Baseline.** The single-polarisation baseline is now Otsu’s (1979) between-class-variance threshold, recomputed independently for every acquisition (Section 3.4.1), directly answering Reviewer 1’s suggestion (specific comment 22) to compare against an established, well-characterised threshold method rather than an ad hoc cut, and it is compared under an identical pipeline to a dual-polarisation SVM that is likewise re-trained per scene (Section 3.4.2).
- **Cost.** A computational-cost audit (Earth Engine Compute Unit usage; Section 3.8, Section 4.4) now accompanies the accuracy comparison, answering Reviewer 1’s question on this point directly.

We address every remaining comment individually below, organised by source (Editor, Reviewer 1, Reviewer 2) and following each reviewer’s own numbering. Where a comment concerned material that no longer appears in the manuscript (chiefly the storage-volume analysis and the multi-classifier comparison), we say so explicitly rather than force an answer to a question that no longer applies.

Response to the Editor

Novelty, scalability, experimental setup. Each is addressed structurally, not just rhetorically: novelty by re-centring the paper on the A/P mechanism (Sections 1, 3.6, 4.1, 5.2); scalability by removing manual training entirely (Section 3.3) and testing on 62 reservoirs on five continents instead of one region (Section 2.1); experimental setup by reorganising Sections 3–4 around the five workflow stages in the workflow figure (Figure 2, as requested by Reviewer 2, items D.1, E.1, E.2) and by removing the storage-volume scope that generated most of the original Results/Discussion difficulties.

Response to Reviewer 1

General comments

1. Workflow is scalable, the model is not. Agreed, and fixed at the root: the model itself no longer requires manual, per-location training. Both detectors are trained automatically from global JRC/WorldCover masks (Section 3.3), and we are explicit that only the *static*, a-priori A/P is shown to generalise; we do *not* claim a dynamic, per-scene switching rule is scalable, because we could not establish that one generalises (Discussion, Section 5.4, final two paragraphs).



2. Only four of many reservoirs validated. The primary comparison now runs on 62 reservoirs referenced against JRC (Sections 2.1, 3.7.1, 4.1–4.2); the four Sicilian reservoirs remain as a secondary, high-resolution near-truth layer (Sections 2.2, 3.7.2, 4.3), not the sole validation.

3. Novelty vs. widely applied SAR+SVM approaches. We agree with the reviewer’s own suggestion: the paper’s contribution is now the quantification of morphometric (A/P) control over SAR classification performance (Sections 1, 3.6, 4.1, 5.2), derived mechanistically from the mixed-pixel argument (Section 1) rather than fitted post hoc to whichever shape metric happened to correlate (Discussion, Section 5.2, final paragraph).

Specific comments

Abstract, “NDWI masks showed strong agreement”: N should be mentioned. Every accuracy figure now states its N explicitly; the Abstract states N=62, and Section 4.2 documents the N=62→59 accounting for the four-detector subset.

Abstract, computational cost of the proposed framework. Added: a dedicated computational-cost audit (Section 3.8, Section 4.4, Figure `eeecu_svm_vs_vv.png`) measuring Earth Engine Compute Unit usage across detector configurations.

Introduction, “relatively few studies... reservoir environments” — novelty of struggling conditions. Rewritten: the Introduction (Section 1) now poses the open question directly (final paragraph before the objectives) rather than asserting that SAR struggles in these conditions as itself the finding.

Introduction, incidence-angle filtering not quantified unlike shoreline complexity. Section 3.2 now states and justifies the orbit-selection rule explicitly (highest mean incidence angle, among orbits with ≥ 20 scenes and $\geq 90\%$ AOI coverage) as a controlled preprocessing choice held constant across both detectors. Shoreline complexity (A/P) remains the sole a-priori predictor under test, consistent with the reviewer’s own assessment that it is the stronger contribution.

Introduction, medium/small reservoirs known to be difficult. The global set spans $\sim 2\text{--}75\text{ km}^2$ (Section 2.1) including medium and small reservoirs, and Section 4.1 shows why: small reservoirs tend toward low A/P, so A/P (not size per se) is the operative geometric control, explaining rather than merely acknowledging the difficulty.

Study Area, units notation (“ $1.12 \times 10^9\text{ m}^3$ ”). No longer applicable: the manuscript no longer reports storage-volume figures (see Summary of the reframing, above); all quantities are surface area (km^2/ha) or shoreline length (m).

Data Collection, “not affected by data lacks due to cloudiness” — how is cloudiness an issue for SAR? Reworded in the Introduction (Section 1) to state plainly that SAR is *not* affected by cloud cover, which is its advantage over optical alternatives, not a claim that cloudiness affects SAR.

Figure 2, text overflow; incidence-angle “why” missing. The workflow figure has been re-generated (`workflow_diagram.pdf`), and Section 3.2 now states the quantitative rationale for the



orbit-selection rule.

Classification, “the labeled dataset” undefined. Fully specified in Section 3.3: water points where JRC occurrence $\geq 95\%$ (relaxed to $\geq 80\%$ where needed), land points from a 500–2000 m ring restricted to non-water WorldCover pixels with zero JRC occurrence, ≤ 500 points per class, fixed random seed. Figure `sample_selection_pozzillo.png` illustrates the construction concretely.

Classifier Selection, “overall accuracy (96.12%)” — train, test, or all? No longer applicable: the manuscript no longer reports a single pixel-level confusion-matrix accuracy. Skill is reported exclusively as KGE (with its r , α , β components) against independent reference data (JRC or PlanetScope) on validation months (Section 3.7), never on the classifier’s own training pixels.

Area Calculation, reservoir boundary / buffered region. Section 3.5 describes the post-processing chain (morphological gap-filling, vectorisation, centroid-inside retention with a largest-polygon fallback). The boundary itself is the JRC maximum-extent polygon (Section 3.3), which by construction already reflects the reservoir’s historical maximum inundation, so it is not a nominal design boundary that could clip real flooding.

Temporal Filtering, LOWESS does not address wind artifacts; mismatch with Introduction. Corrected: Section 3.5 now states plainly that LOWESS smoothing is applied for temporal consistency only, and that accuracy metrics are computed on the raw, unsmoothed classification. Wind’s actual effect is instead tested directly in Sections 3.9/4.5, where we report an honest null result for a systematic wind effect (Discussion, Section 5.3), replacing the earlier, untested claim.

Volume Calculation, 2D polynomial vs. using AEV curves directly. No longer applicable: the manuscript does not attempt area-to-volume conversion (see Summary of the reframing). Discussion Section 5.4 identifies AEV reconstruction as future work, precisely because, as the reviewer rightly noted, that conversion carries separate uncertainties this paper is not positioned to resolve.

Validation, present only the validation/test period. No longer applicable in its original form (there is no train/test split of a pixel classifier). The corresponding safeguard is that KGE (Section 3.7) is computed strictly over months where an independent external reference exists, never over training data.

Results, manual delineation undercuts the scalability claim. Fixed directly: Section 3.3 describes a fully automated sample-selection procedure; no step in the pipeline involves manual delineation.

Figure 6, confusion-matrix broken reference and N mismatch. No longer applicable: confusion-matrix figures have been removed. The four-detector comparison (Figure `kge_4way.png`, Section 4.2) reports paired per-reservoir KGE with an internally consistent, explicitly documented N at every step.

Results, NDWI threshold sentence awkward; justification belongs in main text. Section 3.7.2 now states the PlanetScope NDWI method concisely in the main text (scene-specific threshold, independent of the SAR workflow); full worked detail remains in the Supplementary Material.



Results, topographic shadowing does not fully explain overestimation. No longer applicable in its original form: with the storage-volume analysis removed, this discrepancy no longer appears. Section 4.3 reports the PlanetScope comparison's bias (β) component directly, without attributing it solely to topography.

Figure 12, why not derive area from in-situ storage/AEV instead of PlanetScope? Explained in Sections 2.2/3.7.2: PlanetScope is used specifically because it is independent of both the AEV relationship and the SAR pipeline, avoiding circularity; in-situ storage records would validate a volume step this paper no longer includes.

Table 3, ambiguous column headers (SVM vs. threshold). No longer applicable in the same form: result tables were rebuilt with every detector (VV Otsu / SVM) named explicitly in each table and figure legend (Table 1; Section 4).

Discussion, physical mechanism behind high-storage underestimation not explained. No longer applicable: no storage/volume analysis remains. The analogous geometric mechanism for area-classification error — the mixed-pixel argument — is instead stated mechanistically up front (Section 1; Section 3.6), not inferred post hoc from a discrepancy.

Discussion, “scalability” via a single Brazil AOI example. That single-example transferability claim has been removed. Scalability is now demonstrated, not asserted, by testing the automated pipeline on 62 reservoirs across five continents and 33 countries (Section 2.1), with the explicit caveat (Section 5.4) that only static, a-priori A/P, not any dynamic switching rule, is shown to generalise.

Discussion, compare against Otsu (1979) rather than an ad hoc simple threshold. Implemented directly: Otsu's (1979) threshold, recomputed per scene, is now the single-polarisation baseline and headline comparison partner to the SVM throughout the manuscript (Sections 3.4.1, 4.2, 5.1).

Discussion, VH reduces wind-induced error — evidence? Corrected: Sections 3.9/4.5 test this directly using ERA5 wind speed at each overpass. We find no systematic wind effect on the between-detector divergence and report this null result honestly (Section 5.3), rather than asserting VH's benefit without evidence.

Response to Reviewer 2

A. Novelty. As above: the manuscript's claim is no longer that GEE, VV/VH polarisation, or incidence-angle prioritisation are themselves novel. The contribution is the A/P mechanism (Sections 1, 3.6, 4.1, 5.2).

B.1 Introduction under-referenced / citations not well matched to claims. The Introduction (Section 1) has been substantially rewritten and re-referenced throughout, with every claim tied to a specific citation (IPCC2023; Pokhrel2021; Cammalleri2022; Bevacqua2023; ISpra2023; Biancamaria2016; Schwatke2015; Zhang2026; Lehner2011; Hao2024; Schwatke2020; Torres2012; Mullissa2021; Pekel2016; Gorelick2017; BhatpuriaEtAl2024; ValerioEtAl2024; PengEtAl2025; Bioresita2018; PenaLuque2021; Martinis2015; McQuillan2025).



B.2 Area vs. level claim overlooks morphology-dependence. Reframed: Section 1 now presents water level and surface area as two complementary observables with different revisit and scaling trade-offs, rather than asserting area is generally superior. The morphology-dependence the reviewer raised is, in fact, the paper’s central subject: A/P is precisely what governs how well area itself can be measured (Sections 3.6, 4.1).

B.3 Outdated AEV argument undermines rather than supports the rationale. Reworded (Section 1): AEV relationships needing updates is now presented as an open problem this paper does not solve — motivating Schwatke-style future work (Discussion, Section 5.4) — not as a justification for preferring area over level.

C. Area of Study needs a summary table of reservoir attributes. Added: Table 1 (Section 2.1) lists per-reservoir country, climate zone, surface area, and A/P for all 62 global reservoirs plus the four Sicilian near-truth reservoirs.

D.1 Sections 2.2–2.5 disorganised relative to Figure 2’s workflow. Section 3 has been fully reorganised around the five workflow stages shown in Figure 2 (the workflow diagram): 3.1 overview, 3.2 acquisition/preprocessing, 3.3 automated training, 3.4 the two detectors, 3.5 post-processing/area, 3.6 morphometry, 3.7 validation, 3.8 cost audit, 3.9 radiometric/climate limits — a direct one-to-one match to the workflow figure.

D.2 IW3 / “when necessary” needs clarification. Section 3.2 now states the orbit-selection rule explicitly and its rationale (see Reviewer 1, specific comment on incidence-angle filtering, above).

D.3 Multiple classifiers (SVM, RF, CART, GTB, Minimum Distance) under-described. The comparison has been narrowed to exactly two, thoroughly described detectors — Otsu (Section 3.4.1) and SVM (Section 3.4.2, with kernel, cost, and γ stated explicitly) — removing the under-justified multi-algorithm comparison entirely.

D.4 Equation (1), the 10^4 factor unexplained. No longer applicable: area is now stated directly as the pixel-area sum of the retained water mask (Section 3.5), without the equation the reviewer flagged.

D.5 Justify the representativeness of the four validation reservoirs. Section 2.2: the four Sicilian reservoirs are now explicitly an additional near-truth PlanetScope tier layered on the 62-reservoir global JRC tier (not the sole validation), chosen to span contrasting morphometries and A/P values within Sicily (Figure `sicily_4lakes_3d.png`) rather than to represent the full global study area by themselves.

E.1 Data-collection subsection misplaced in Results. Fixed: all data and methods content now resides in Section 3 only; Section 4 (Results) contains no data-collection description.

E.2 Results section reorganisation. Section 4 is restructured into five subsections matching the paper’s actual claims (4.1 A/P a priori, 4.2 four-way detector comparison, 4.3 PlanetScope validation, 4.4 cost, 4.5 wind/climate), no longer interleaved with workflow-step descriptions.



E.3 Hyperparameter tuning (three values of C and γ) unclear. No longer applicable: the SVM now uses a single, fixed, fully stated configuration ($C = 1$, $\gamma = 0.01$; Section 3.4.2) rather than an ambiguously reported grid search.

E.4 Storage discrepancy interpretation inconsistent with sedimentation direction. No longer applicable: the storage-volume analysis has been removed in its entirety (see Summary of the reframing), so the disputed claim no longer appears.

E.5 Only four of 41 reservoirs shown; extend further. Addressed by extending well beyond the original 41-reservoir Sicilian study area to 62 reservoirs on five continents as the primary comparison (Section 2.1), with the four Sicilian reservoirs retained only as an additional near-truth layer (Section 2.2).

F. Robustness/generalisability across all 41 reservoirs unclear. Representativeness is now argued at the 62-reservoir, five-continent, A/P-stratified level (Section 2.1, Figure `study_area_map`), not inferred from four of 41 Sicilian reservoirs alone.

We believe these changes directly answer the concerns raised by the Editor and both Reviewers, and we hope the manuscript is now suitable for review at *Environmental Modelling & Software*. We would be glad to provide any further clarification.

Sincerely,

Cicero Martins Jr.

on behalf of all co-authors